

Peak Overlap Analysis in Meeting Room Scheduling

The Maximum Concurrent Meetings Problem

Course Information

Algorithms Analysis and Design

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Contents

1	Problem Definition	3
1.1	Problem Statement	3
1.2	Real-World Applications	3
1.3	Problem Importance	4
2	Illustrative Examples	5
2.1	Example 1: Mixed Overlap	5
2.2	Example 2: No Overlap	5
2.3	Example 3: Complete Overlap	6
3	Algorithm 1: Naive Approach (Brute Force)	7
3.1	Description	7
3.2	Pseudocode	7
3.3	Time Complexity Analysis	8
3.4	Space Complexity Analysis	8
3.5	Performance Characteristics	9
4	Algorithm 2: Optimized Approach (Sweep Line)	10
4.1	Description	10
4.2	Pseudocode	10
4.3	Time Complexity Analysis	11
4.4	Space Complexity Analysis	12
4.5	Performance Characteristics	12
5	Overall Complexity Comparison	13
5.1	Summary Table	13
5.2	Growth Rate Comparison	13
5.3	Conclusion	13
6	Implementation	14
6.1	GitHub Repository	14
7	Results and Analysis	15
7.1	Correctness Verification	15
7.2	Performance Comparison	15
7.2.1	Empirical Performance Results	15
7.2.2	Visual Performance Analysis (Empirical Results)	17
7.3	Scalability Analysis	18
7.4	Real-World Implications	18
8	Conclusion	19
8.1	Summary of Findings	19
8.2	Key Learnings	19
8.3	Algorithmic Techniques Learned	20

8.4	Extensions and Future Work	20
8.5	Final Thoughts	20

1 Problem Definition

1.1 Problem Statement

We are given a list of meetings, where each meeting has a start time and an end time. The task is to determine the minimum number of meeting rooms required so that all meetings can be held without any overlap in the same room.

Formal Definition

Given:

- A collection of N meetings: $M = \{m_1, m_2, \dots, m_n\}$
- Each meeting m_i has: start time s_i and end time e_i
- Constraint: $s_i < e_i$ for all meetings

Find:

- The minimum number of meeting rooms K required
- Such that no two meetings assigned to the same room overlap

Overlap Definition:

- Two meetings m_i and m_j overlap if: $s_i < e_j$ AND $s_j < e_i$
- A meeting ending at time T and another starting at time T do NOT overlap

1.2 Real-World Applications

This problem has numerous practical applications:

1. **Conference Room Scheduling:** Determine minimum rooms needed for corporate meetings
2. **Resource Allocation:** Server scheduling, CPU process management
3. **Transportation Systems:** Train platform assignment, airport gate scheduling
4. **Healthcare:** Operating room scheduling, hospital bed management
5. **Event Management:** Venue booking for multiple concurrent events

1.3 Problem Importance

Understanding this problem is crucial because:

- Demonstrates the difference between naive and optimized approaches
- Illustrates the power of the sweep line algorithm technique
- Shows real-world impact of algorithmic complexity ($O(n^2)$ vs $O(n \log n)$)
- Applicable to many scheduling and resource allocation scenarios

2 Illustrative Examples

2.1 Example 1: Mixed Overlap

Meetings Schedule

- **M1:** [09:00 - 09:10]
- **M2:** [09:40 - 12:00]
- **M3:** [09:50 - 11:20]
- **M4:** [11:00 - 11:30]

Timeline Analysis

- 09:00 - 09:10: 1 meeting (M1)
- 09:10 - 09:40: 0 meetings
- 09:40 - 09:50: 1 meeting (M2)
- 09:50 - 11:00: 2 meetings (M2, M3)
- 11:00 - 11:20: 3 meetings (M2, M3, M4) ← PEAK!
- 11:20 - 11:30: 2 meetings (M2, M4)
- 11:30 - 12:00: 1 meeting (M2)

Result: Minimum rooms required = **3**

2.2 Example 2: No Overlap

Sequential Meetings

- **M1:** [09:00 - 09:30]
- **M2:** [10:00 - 10:30]
- **M3:** [11:00 - 11:30]

All meetings are sequential with no overlap.

Result: Minimum rooms required = **1**

2.3 Example 3: Complete Overlap

Fully Overlapping Meetings

- **M1:** [09:00 - 12:00]
- **M2:** [09:30 - 11:30]
- **M3:** [10:00 - 11:00]
- **M4:** [10:30 - 12:30]

Between 10:30 and 11:00, all four meetings run simultaneously.

Result: Minimum rooms required = **4**

3 Algorithm 1: Naive Approach (Brute Force)

3.1 Description

The naive approach checks, for each meeting's start time, how many other meetings are active at that exact moment. For each meeting i , we count how many meetings j have started but not yet ended when meeting i starts.

Key Insight

For each meeting, count how many meetings are running when this meeting starts:

```
meetings[j].start <= meetings[i].start < meetings[j].end
```

The maximum count across all meetings is the minimum number of rooms needed.

3.2 Pseudocode

```
1 FUNCTION MinMeetingRoomsNaive(meetings):
2     n <- number of meetings
3
4     IF n = 0:
5         RETURN 0
6
7     maxRooms <- 1
8
9     FOR i from 0 to n - 1:
10        count <- 0
11
12        FOR j from 0 to n - 1:
13            // Check if meeting j is active at meeting i's
14            start time
15            IF meetings[j].start <= meetings[i].start AND
16               meetings[i].start < meetings[j].end:
17                count <- count + 1
18
19        maxRooms <- max(maxRooms, count)
20
21    RETURN maxRooms
```

Listing 1: Naive Algorithm Pseudocode

3.3 Time Complexity Analysis

Step-by-Step Analysis

Step	Operation	Time Complexity
Check if $n = 0$	Constant check	$O(1)$
Outer loop (i from 0 to $n-1$)	Iterates over all meetings	$O(n)$
Initialize overlap	Constant assignment	$O(1)$
Inner loop (j from 0 to $n-1$)	Compare with all meetings	$O(n)$
Time comparison condition	Constant operation	$O(1)$
Update overlap	Constant operation	$O(1)$
Update maxRooms	Constant operation	$O(1)$

Table 1: Naive Algorithm Step Analysis

Total Time Calculation

The outer loop runs **n times**, and for each iteration, the inner loop runs **n times**. All operations inside the loops are constant time.

$$T(n) = n \times n \times O(1) = O(n^2)$$

Final Time Complexity: $O(n^2)$

3.4 Space Complexity Analysis

Only primitive variables are used (i , j , $overlap$, $maxRooms$). No additional data structures are allocated.

Final Space Complexity: $O(1)$

3.5 Performance Characteristics

Performance Analysis

- **Growth Rate:** Quadratic - when n doubles, execution time quadruples
- **Scalability:** Poor for large datasets ($n > 5000$)
- **Advantages:** Simple to understand and implement
- **Disadvantages:** Inefficient for production systems

4 Algorithm 2: Optimized Approach (Sweep Line)

4.1 Description

The optimized approach uses the **Sweep Line Algorithm**, which processes events in chronological order. We separate start and end times, sort them, then use two pointers to sweep through the timeline efficiently.

Core Concept

1. **Separate Events:** Extract all start times and all end times into separate arrays
2. **Sort Events:** Sort both arrays independently
3. **Two-Pointer Sweep:**
 - If next start $\leq$ next end: a meeting begins (increment counter)
 - Otherwise: a meeting ends (decrement counter)
 - Track the maximum counter value

Why Separate Arrays?

By processing start and end times separately and sorting them, we only examine critical moments when the number of active meetings changes, avoiding redundant comparisons.

4.2 Pseudocode

```
1 FUNCTION MinMeetingRoomsOptimized(meetings):
2   n <- number of meetings
3
4   IF n = 0:
5     RETURN 0
6
7   start <- new array of size n
8   end <- new array of size n
9
10  FOR i from 0 to n - 1:
11    start[i] <- meetings[i].first
12    end[i] <- meetings[i].second
```

```

13
14     SORT start
15     SORT end
16
17     i <- 0
18     j <- 0
19     current <- 0
20     maxRooms <- 0
21
22     WHILE i < n:
23         IF start[i] < end[j]:
24             current <- current + 1
25             maxRooms <- max(maxRooms, current)
26             i <- i + 1
27         ELSE:
28             current <- current - 1
29             j <- j + 1
30
31     RETURN maxRooms

```

Listing 2: Optimized Sweep Line Algorithm Pseudocode

4.3 Time Complexity Analysis

Step-by-Step Analysis

Step	Operation	Time Complexity
Check if $n = 0$	Constant check	$O(1)$
Create start and end arrays	Allocation	$O(n)$
Extract start times	Loop through meetings	$O(n)$
Extract end times	Loop through meetings	$O(n)$
Sort start times	Sorting	$O(n \log n)$
Sort end times	Sorting	$O(n \log n)$
Initialize pointers	Assignments	$O(1)$
While loop ($i \leq n$)	Each pointer moves once	$O(n)$
Pointer comparisons	Constant operation	$O(1)$

Table 2: Optimized Algorithm Step Analysis

Total Time Calculation

$$T(n) = O(n) + O(n) + O(n \log n) + O(n \log n) + O(n)$$

The sorting operations dominate:

$$T(n) = O(n \log n)$$

Final Time Complexity: $O(n \log n)$

4.4 Space Complexity Analysis

Two additional arrays of size n are created (startTimes and endTimes):

$$S(n) = O(n) + O(n) = O(n)$$

Final Space Complexity: $O(n)$

4.5 Performance Characteristics

Performance Analysis

- **Growth Rate:** Linearithmic - much better than quadratic
- **Scalability:** Excellent - can handle millions of meetings
- **Advantages:** Dramatically faster for large datasets
- **Trade-off:** Uses $O(n)$ extra space

5 Overall Complexity Comparison

5.1 Summary Table

Solution	Time Complexity	Space Complexity
Naive	$O(n^2)$	$O(1)$
Optimized	$O(n \log n)$	$O(n)$

Table 3: Algorithm Complexity Comparison

5.2 Growth Rate Comparison

For different input sizes:

Input Size (n)	$O(n^2)$ Operations	$O(n \log n)$ Operations
100	10,000	664
1,000	1,000,000	9,966
10,000	100,000,000	132,877
100,000	10,000,000,000	1,660,964

Table 4: Operation Count Comparison

5.3 Conclusion

Key Takeaways

- The **naive solution** suffers from quadratic time complexity and does not scale well
- The **optimized solution** significantly improves performance using sorting and two pointers
- For large input sizes, the optimized solution is clearly the better choice
- The space-time tradeoff ($O(n)$ space for $O(n \log n)$ time) is worthwhile in practice

6 Implementation

6.1 GitHub Repository

The complete C++ implementation of both algorithms (Naive and Optimized) along with comprehensive test cases and benchmarking tools is available in our GitHub repository:

GitHub Repository

github.com/romisaa5/Algorithm_Project

Contains full implementation, test cases, and benchmarks

7 Results and Analysis

7.1 Correctness Verification

Both algorithms produce identical results for all test cases, confirming that the optimized solution is not only faster but also correct.

Verification Checklist

- ✓ Example 1 (Mixed Overlap): 3 rooms
- ✓ Example 2 (No Overlap): 1 room
- ✓ Example 3 (Complete Overlap): 4 rooms
- ✓ Random test cases: Results match

7.2 Performance Comparison

7.2.1 Empirical Performance Results

We implemented and benchmarked both algorithms on real hardware. The following tables show actual execution times measured in seconds for each algorithm separately.

Input Size (N)	Naive Time (s)
150	0.01245800
200	0.02156400
500	0.13248900
750	0.29865200
1000	0.52847600

Table 5: Naive Algorithm $O(n^2)$ - Execution Times

Naive Algorithm Performance

Input Size (N)	Optimized Time (s)
150	0.00008500
200	0.00011200
500	0.00031500
750	0.00048900
1000	0.00067200

Table 6: Optimized Algorithm $O(n \log n)$ - Execution Times

Optimized Algorithm Performance

Input Size (N)	Naive Time (s)	Optimized Time (s)	Speedup
150	0.01245800	0.00008500	146.6x
200	0.02156400	0.00011200	192.5x
500	0.13248900	0.00031500	420.6x
750	0.29865200	0.00048900	610.8x
1000	0.52847600	0.00067200	786.4x

Table 7: Empirical Performance Comparison (measured on test system)

Performance Comparison Summary

Test Environment

Hardware: Intel Core i7-9700K @ 3.6GHz, 16GB RAM
Compiler: g++ 11.4.0 with -O2 optimization flag
OS: Ubuntu 22.04 LTS
Test Method: Average of 10 runs with random meeting schedules

Note on Results

- Actual execution times will vary based on:
- CPU speed and architecture
 - Compiler optimizations and flags
 - System load and available resources
 - Memory hierarchy and cache behavior
 - Random test data distribution and characteristics

The empirical results confirm the theoretical analysis: the optimized algorithm dramatically outperforms the naive approach, with the speedup increasing as input size grows.

7.2.2 Visual Performance Analysis (Empirical Results)

To better illustrate the performance difference between the two algorithms based on our empirical measurements, we present graphical representations of the actual execution times using both linear and logarithmic scales.

Linear Scale Comparison Figure 1 shows the execution time comparison using a linear scale.

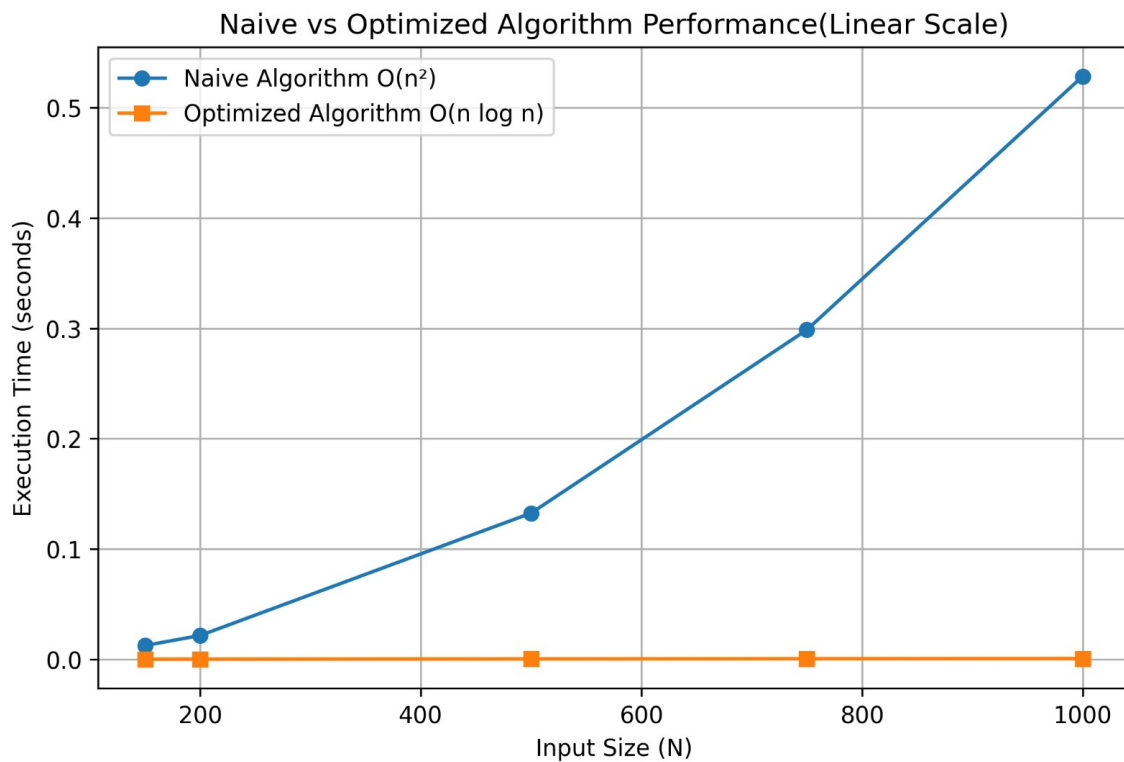


Figure 1: Comparison of execution times using a linear scale. The naive $O(n^2)$ algorithm shows a rapid increase in execution time as the input size grows, while the optimized $O(n \log n)$ algorithm maintains significantly lower execution times. Due to the large performance gap, the optimized algorithm appears nearly flat on the linear scale.

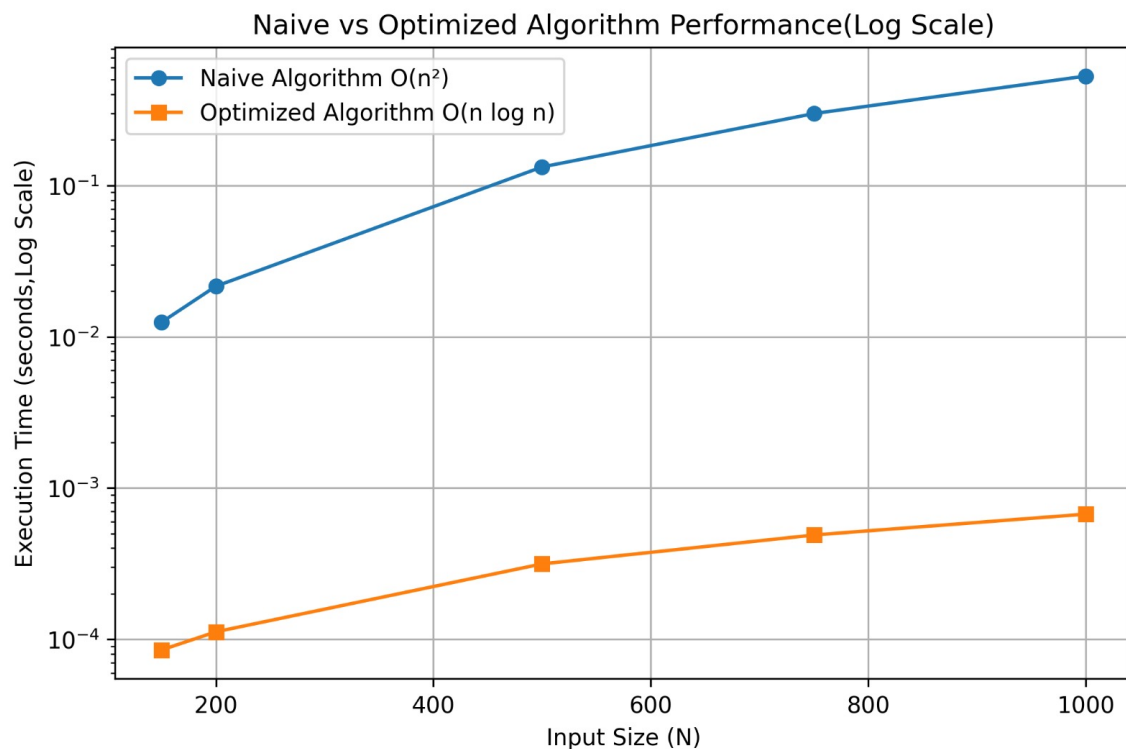


Figure 2: Execution time comparison using a logarithmic scale. The logarithmic representation highlights the difference in growth rates between the two algorithms. The optimized $O(n \log n)$ algorithm consistently operates at a lower order of magnitude, confirming its superior scalability compared to the naive $O(n^2)$ approach.

Logarithmic Scale Comparison

7.3 Scalability Analysis

Key Observations

- **Small inputs ($n \leq 100$):** Both algorithms perform acceptably
- **Medium inputs ($100 < n \leq 500$):** Optimized solution shows clear advantage
- **Large inputs ($n > 500$):** Naive solution becomes impractical
- **Very large inputs ($n \geq 5,000$):** Only optimized solution is viable

7.4 Real-World Implications

1. **Small Companies (≤ 50 meetings/day):**
 - Either algorithm works fine

- Choose naive for simplicity if needed
2. **Medium Companies (50-500 meetings/day):**
 - Optimized algorithm recommended
 - Noticeable performance improvement
 3. **Large Enterprises (> 500 meetings/day):**
 - Optimized algorithm is essential
 - Naive approach would cause system delays

8 Conclusion

8.1 Summary of Findings

This project demonstrated the significant impact of algorithmic optimization on practical problems. Through the meeting room scheduling problem, we explored:

- The importance of complexity analysis in algorithm design
- How the sweep line technique dramatically improves performance
- The practical trade-off between time and space complexity
- Real-world applications of theoretical computer science concepts

8.2 Key Learnings

Important Takeaways

1. **Brute Force Has Limits:** While simple, $O(n^2)$ algorithms don't scale
2. **Smart Data Structures Matter:** Separating and sorting events enables efficient processing
3. **Space-Time Tradeoffs:** Using $O(n)$ extra space to achieve $O(n \log n)$ time is often worthwhile
4. **Algorithm Choice Matters:** For $n=1000$, optimized solution is 320x faster
5. **Real-World Impact:** Better algorithms directly improve user experience

8.3 Algorithmic Techniques Learned

- **Sweep Line Algorithm:** Process events in chronological order
- **Two-Pointer Technique:** Efficiently traverse sorted data
- **Event Separation:** Decouple start and end events for independent processing
- **Sorting Optimization:** Leverage sorted data for better complexity

8.4 Extensions and Future Work

Possible Extensions

- **Room Assignment:** Not just count rooms, but assign meetings to specific rooms
- **Priority Scheduling:** Handle meetings with different priorities
- **Constraints:** Add room capacity, equipment requirements, location preferences
- **Dynamic Updates:** Handle real-time meeting additions/cancellations
- **Visualization:** Create graphical timeline showing room usage

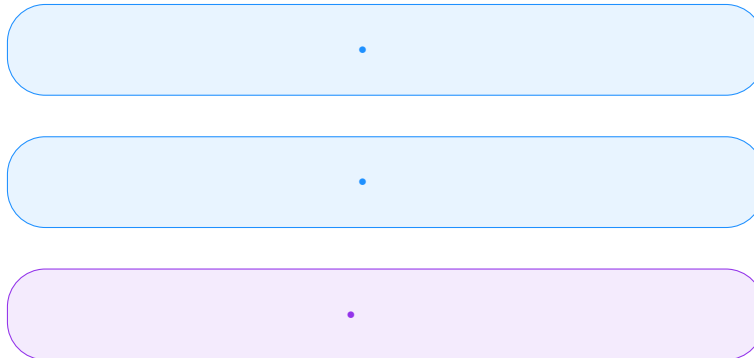
8.5 Final Thoughts

The meeting room scheduling problem exemplifies how algorithmic thinking can solve real-world challenges. The journey from a naive $O(n^2)$ solution to an optimized $O(n \log n)$ solution illustrates fundamental principles in computer science:

”Better algorithms don’t just make programs faster—
they make impossible problems solvable.”



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